



DR. S. RAMANATHAN

Team Lead – Machine Vision | Ph.D. Control Systems

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LINKS

LINKEDIN

ramanathan-s-8338a33b3

SCHOLAR

scholar.google.com
(TmbVeDMAAAAJ)

PORTFOLIO

ramgom1041.github.io/ramanathan-site

PERSONAL

DATE OF BIRTH

02 May 1994

NATIONALITY

Indian

LANGUAGES

Tamil, English

MARITAL STATUS

Single

CORE SKILLS

- Machine Vision Systems
- Control Systems & PID
- Power Electronics
- PCB Design – KiCAD, Proteus
- MATLAB / Simulink
- LabVIEW
- PLC – Delta, Omron
- Xilinx FPGA
- Instrumentation & Calibration

PROFICIENCY

ADVANCED

LabVIEW, MATLAB, Delta PLC

INTERMEDIATE

KiCAD, Arduino

BEGINNER

Xilinx

PROFESSIONAL SUMMARY

Ph.D.-qualified engineering professional with 8+ years of combined industry, research, and academic experience spanning machine vision, control systems, power electronics, instrumentation, and renewable energy. Currently leading the Machine Vision team at Manatec Electronics Pvt. Ltd., directing the design, development, and validation of a 3D geometry measurement system. Published author in IEEE Transactions on Industrial Electronics with a proven record of converting advanced research into reliable, production-ready engineering solutions.

PROFESSIONAL EXPERIENCE

Team Lead – Machine Vision

Sep 2025 – Present

Manatec Electronics Pvt. Ltd., Puducherry

- Lead a 4-member Machine Vision team across design, development, and testing of vision-based inspection and measurement systems.
- Drive the implementation of a 3D geometry measurement system, from concept through validation.

Lead – Product Development

Oct 2024 – Aug 2025

Lectionic Solutions Pvt. Ltd., Puducherry

- Designed the PCB for a tank water-level controller (circuit design + layout in KiCAD).
- Designed a pressure transmitter and developed its Windows-based calibration interface application.
- Executed PCB testing and hardware validation to ensure design integrity.

Doctoral Researcher

2018 – 2024

Puducherry Technological University (PEC)

- Designed a nonlinear controller to stabilize DC-link voltage and protect power electronics during 3 ϕ grid faults in DFIG-based wind turbines.
- Validated across 2 MW simulations and 0.75 kW hardware, improving fault tolerance and reducing voltage fluctuations.
- Authored 1 IEEE Transactions journal article and 2 international conference papers.

Guest Faculty – Dept. of EIE

2019 – 2020

Pondicherry Engineering College

- Delivered Electronics & Instrumentation Engineering courses to undergraduate students.

Assistant Professor – Biomedical Engg.

2017 – 2018

Rajiv Gandhi College of Engineering & Technology

- Taught undergraduate Biomedical Engineering courses.

EDUCATION

Ph.D., Electrical Engineering (Control Systems)

2018 – 2024

Pondicherry University

M.Tech, Instrumentation Engineering – CGPA 8.19

2015 – 2017

Pondicherry University

B.Tech, Electronics & Instrumentation Engineering – CGPA 6.71

2012 – 2015

Pondicherry University

Diploma, Instrumentation & Control Engineering – 83.21%

2009 – 2012

MN Govt. Polytechnic, Lawspet

PUBLICATIONS

- S. Ramanathan et al., “Nonlinear Control Strategy for DC-Link Voltage Control in a DFIG of WECS During 3 ϕ

Grid Faults,” IEEE Transactions on Industrial Electronics, 2024. doi:10.1109/TIE.2023.3344815.

2. S. Ramanathan et al., “Indirect Power Control MPPT Techniques for DFIG-based WEGs: A Comparative Study,” ICSECT-2022, Pandit Deendayal Energy University, Gandhinagar.
3. S. Ramanathan et al., “Operation of DFIG-based WECS under Symmetrical Voltage Dip using Crowbar,” ICEEMR-2022, Chennai Institute of Technology, pp. 250–259.
4. B. Hema Kumar, S. Ramanathan, “A Novel Smart Highway Automated Toll Collection System using Raspberry Pi & OpenCV,” Int. J. of Engineering Research and Applications (IJERA), 8(5), pp. 81–85, 2018.
5. S. Ramanathan, B. Hemakumar, B. Varadharaju, “A Virtual Instrument Approach to Model Jaw Tremor,” Australian J. of Basic and Applied Sciences, 11(11), pp. 148–155, 2017.
6. S. Ramanathan, “Wiener-based PI Controller for Strong Acid & Strong Base pH Neutralization Process” (oral presentation), TIMA-2017, MIT Campus, Anna University, Chennai.

PROFESSIONAL DEVELOPMENT

- AICTE ATAL Academy FDPs — Energy Engineering (NIT Tiruchirappalli); EV Charging Technology in Smart Grid (PTU); Renewable-Energy-Based Microgrid Systems (NIT Mizoram); IoT in Renewable Energy (NSS CE); Green Technology & Sustainability Engineering (NSUT).
- AICTE-ISTE Orientation/Refresher Programme — Physiological Control & Internet of Medical Things (PTU, 2021).
- Workshops & Training — Nuclear Control & Instrumentation (SPARC, NIT Trichy); System Identification (NWOSI, PEC); MEMS Design Tool – IntelliSuite; System Design using Raspberry Pi; Multivariate Data Analysis (Grade A); Cybersecurity (ICECON-2019); Research Rejuvenation (TEQIP-III).
- Seminar — Advancement in Surgical Instruments & Bio-signal Processing (ICMR-sponsored, RGCT, 2017).

REFERENCES

Prof. R. Ananda Natarajan

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Prof. P. Ramesh Babu

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